PAC-Bayes Generalization Bound for the SCX Noise Detector

Paper 3 Upgrade Item 1 Status: Draft — proof complete modulo standard PAC-Bayes machinery **Target location**: Paper 3 Section 4.2 (Theorem 5) **Date**: 2026-06-27

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1. Motivation

Current limitation of Theorem 1: Theorem 1 bounds the true F1 assuming knowledge of the state-level expert error rates $\{\mu_s\}_{s\in\mathcal{S}}$:

$$\mathrm{F1} \geq 1 - \frac{1}{\eta}\sum_{s\in\mathcal{S}} \rho_s \cdot \exp\!\left(-2M\Delta_s^2\right)$$

where $\Delta_s = \min(\theta - \mu_s,\ 1 - C_{\mathrm{bal}} \cdot \mu_s/(K-1) - \theta)$. In practice, μ_s is **unknown** and must be estimated from a finite clean validation set. The threshold θ is then chosen based on the estimates $\{\hat{\mu}_s\}$.

The problem: When we plug $\hat{\mu}_s$ into Theorem 1, the resulting bound is an *estimate* of the true F1. We need a bound that holds with high probability over the randomness of the validation set, accounting for the complexity of the threshold selection procedure.

The solution: PAC-Bayes theory (McAllester, 1999; Catoni, 2007) provides a principled way to bound the gap between empirical and true performance when the hypothesis (here: the detection threshold) is chosen in a data-dependent manner. The penalty depends on the KL divergence between a data-dependent posterior and a fixed prior, measuring the "information cost" of adapting the threshold to the data.

Value: This makes the F1 bound fully empirical — no oracle quantities required.

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2. Setup and Notation

2.1 The Noise Detector as a Classifier

The SCX noise detector $h_\theta : \mathcal{X} \to \{0, 1\}$ is a binary classifier:

$$h_\theta(x, y, \{f_m\}_{m=1}^M) = \mathbf{1}\{C(x) > \theta\}, \quad C(x) = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\{\ell(f_m(x), y) > \tau\}$$

where output 1 means "predicted noise" and 0 means "predicted clean."

2.2 Loss Function

Let $z \in \{0, 1\}$ be the true noise indicator ($z = 1$ means the label is corrupted). Define the **detection loss**:

$$\ell_\theta(x, y, z) = \mathbf{1}\{h_\theta(x) \neq z\}$$

This is the 0-1 classification loss for the noise detection task. Its expectation is the **detection error rate**:

$$R(\theta) = \mathbb{E}[\ell_\theta(X, Y, Z)] = \eta \cdot \mathrm{FNR}(\theta) + (1 - \eta) \cdot \mathrm{FPR}(\theta)$$

2.3 Relation to F1 Score

The F1 score is connected to $R(\theta)$ through the precision-recall tradeoff. Specifically:

$$\mathrm{F1}(\theta) = \frac{2\eta(1 - \mathrm{FNR}(\theta))}{2\eta(1 - \mathrm{FNR}(\theta)) + (1 - \eta)\mathrm{FPR}(\theta) + \eta \cdot \mathrm{FNR}(\theta)}$$

For small FPR and FNR (the regime of interest in Theorem 1), we have the approximation:

$$1 - \mathrm{F1}(\theta) \approx \frac{(1 - \eta)\mathrm{FPR}(\theta) + \eta \cdot \mathrm{FNR}(\theta)}{\eta} = \frac{R(\theta)}{\eta}$$

More precisely, we have the following inequality (derived in Lemma 4 below):

$$1 - \mathrm{F1}(\theta) \leq \frac{R(\theta)}{\eta \cdot (1 - R(\theta))}$$

For $R(\theta) \leq 1/2$ (which holds under Theorem 1 conditions), this simplifies to:

$$1 - \mathrm{F1}(\theta) \leq \frac{2R(\theta)}{\eta}$$

2.4 Validation Set

We have a labeled validation set:

$$V = \{(x_i, y_i, z_i)\}_{i=1}^N \sim \mathcal{D}^N$$

where $z_i = \mathbf{1}\{y_i \neq y_i^*\}$ is the true noise indicator. This validation set may be: - **Synthetic**: Take clean data, inject known noise at rate η - **Natural**: A held-out set with ground-truth labels (e.g., from a domain expert)

The empirical detection error is:

$$\hat{R}_V(\theta) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{h_\theta(x_i, y_i) \neq z_i\}$$

2.5 PAC-Bayes Framework

Let $\Theta \subseteq [0, 1]$ be the hypothesis class of thresholds. Let P be a **prior distribution** over Θ (chosen before seeing V). Let Q be a **posterior distribution** over Θ (possibly data-dependent, e.g., concentrated around an empirically optimal threshold).

The expected risk under Q is:

$$R(Q) = \mathbb{E}_{\theta \sim Q}[R(\theta)], \quad \hat{R}_V(Q) = \mathbb{E}_{\theta \sim Q}[\hat{R}_V(\theta)]$$

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3. Main Theorem

Theorem 5 (PAC-Bayes SCX Generalization)

Let the setup of Section 2 hold. Let P be any prior distribution over Θ that does not depend on V. Assume the loss $\ell_\theta \in [0,1]$ (bounded). Then for any $\delta \in (0,1)$, with probability at least $1-\delta$ over draws of $V \sim \mathcal{D}^N$, for **all** distributions Q over Θ (possibly data-dependent):

$$R(Q) \leq \hat{R}_V(Q) + \sqrt{\frac{\mathrm{KL}(Q \parallel P) + \log \frac{2\sqrt{N}}{\delta}}{2N}}$$

where $\mathrm{KL}(Q \parallel P) = \int \log \frac{dQ}{dP}\, dQ$ is the Kullback-Leibler divergence.

Corollary 5.1 (F1 Form)

Under the same conditions, if $R(\theta) \leq 1/2$ for all $\theta \in \Theta$ (which holds when $\Delta_s > 0$, i.e., under B3), then with probability at least $1-\delta$:

$$1 - \mathrm{F1}(Q) \leq \frac{2}{\eta}\left[\hat{R}_V(Q) + \sqrt{\frac{\mathrm{KL}(Q \parallel P) + \log \frac{2\sqrt{N}}{\delta}}{2N}}\right]$$

Equivalently:

$$\mathrm{F1}(Q) \geq \widehat{\mathrm{F1}}(Q) - \frac{2}{\eta}\sqrt{\frac{\mathrm{KL}(Q \parallel P) + \log \frac{2\sqrt{N}}{\delta}}{2N}}$$

where $\widehat{\mathrm{F1}}(Q) = \mathbb{E}_{\theta\sim Q}[\widehat{\mathrm{F1}}_V(\theta)]$ is the expected empirical F1 on the validation set.

Corollary 5.2 (Threshold Learning Bound)

For the special case where $Q = \delta_{\hat{\theta}}$ is a Dirac at the empirically optimal threshold $\hat{\theta} = \arg\min_\theta \hat{R}_V(\theta)$, and P is a uniform prior over Θ, we have $\mathrm{KL}(\delta_{\hat{\theta}} \parallel P) = \log|\Theta|$ (where $|\Theta|$ is the number of candidate thresholds in a discretized grid). Then:

$$\mathrm{F1}(\hat{\theta}) \geq \widehat{\mathrm{F1}}_V(\hat{\theta}) - \frac{2}{\eta}\sqrt{\frac{\log|\Theta| + \log\frac{2\sqrt{N}}{\delta}}{2N}}$$

This shows that the penalty for threshold selection grows only logarithmically in the number of candidate thresholds — consistent with the Vapnik-Chervonenkis theory for threshold classifiers.

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4. Proof

4.1 Lemma 4 (F1-Risk Relation)

Lemma 4. For any noise detector h with detection error rate $R = \mathbb{P}(h(X) \neq Z)$ and F1 score F1:

$$1 - \mathrm{F1} \leq \frac{R}{\eta \cdot (1 - R)}$$

If $R \leq 1/2$, then:

$$1 - \mathrm{F1} \leq \frac{2R}{\eta}$$

Proof. Recall the definitions:

$$\begin{aligned}
&\text{TPR} = \mathbb{P}(h = 1 \mid Z = 1), \quad \text{FPR} = \mathbb{P}(h = 1 \mid Z = 0), \\
&\text{FNR} = 1 - \text{TPR} = \mathbb{P}(h = 0 \mid Z = 1)
\end{aligned}$$

The detection error rate is:

$$R = \eta \cdot \text{FNR} + (1 - \eta) \cdot \text{FPR}$$

F1 is:

$$\text{F1} = \frac{2\eta \cdot \text{TPR}}{2\eta \cdot \text{TPR} + (1 - \eta) \cdot \text{FPR} + \eta \cdot \text{FNR}}$$

Substituting $\text{TPR} = 1 - \text{FNR}$:

$$\begin{aligned}
\text{F1} &= \frac{2\eta(1 - \text{FNR})}{2\eta(1 - \text{FNR}) + (1 - \eta)\text{FPR} + \eta \cdot \text{FNR}} \\
&= \frac{2\eta - 2\eta \cdot \text{FNR}}{2\eta - 2\eta \cdot \text{FNR} + (1 - \eta)\text{FPR} + \eta \cdot \text{FNR}} \\
&= \frac{2\eta - 2\eta \cdot \text{FNR}}{2\eta - \eta \cdot \text{FNR} + (1 - \eta)\text{FPR}}
\end{aligned}$$

Therefore:

$$\begin{aligned}
1 - \mathrm{F1} = 1 - \frac{2\eta - 2\eta \cdot \mathrm{FNR}}{2\eta - \eta \cdot \mathrm{FNR} + (1-\eta)\mathrm{FPR}} \\
&= \frac{(1-\eta)\mathrm{FPR} + \eta \cdot \mathrm{FNR}}{2\eta - \eta \cdot \mathrm{FNR} + (1-\eta)\mathrm{FPR}} \\
&= \frac{R}{2\eta - \eta \cdot \mathrm{FNR} + (1-\eta)\mathrm{FPR}} \\
&\leq \frac{R}{\eta(2 - \mathrm{FNR})} \quad (\text{since } (1-\eta)\mathrm{FPR} \geq 0) \\
&\leq \frac{R}{\eta} \quad (\text{since } \mathrm{FNR} \leq 1 \Rightarrow 2 - \mathrm{FNR} \geq 1)
\end{aligned}$$

For the refined bound, note that $R = \eta \cdot \mathrm{FNR} + (1-\eta) \cdot \mathrm{FPR}$ and $2\eta - \eta \cdot \mathrm{FNR} + (1-\eta)\mathrm{FPR} = \eta(2 - \mathrm{FNR}) + (1-\eta)\mathrm{FPR}$. Since $2 - \mathrm{FNR} \geq 1$, the denominator is at least η. Moreover, when $R \leq 1/2$, we have $1 - R \geq 1/2$, so:

$$1 - \mathrm{F1} = \frac{R}{2\eta - \eta \cdot \mathrm{FNR} + (1-\eta)\mathrm{FPR}} \leq \frac{R}{2\eta - \eta \cdot \mathrm{FNR}} \leq \frac{R}{\eta}$$

The tighter bound $1 - \mathrm{F1} \leq R/(\eta \cdot (1 - R))$ follows from noting that:

$$\begin{aligned}
2\eta - \eta \cdot \mathrm{FNR} + (1-\eta)\mathrm{FPR} &= \eta(1 + \mathrm{TPR}) + (1-\eta)\mathrm{FPR} \\
&\geq \eta(1 + \mathrm{TPR}) \geq \eta(1 + 1 - \mathrm{FNR}) = \eta(2 - \mathrm{FNR})
\end{aligned}$$

And since $R \geq \eta \cdot \mathrm{FNR}$:

$$2 - \mathrm{FNR} \geq 2 - \frac{R}{\eta} \geq 1 - R$$

when $R \leq \eta \leq 1/2$. In general, $1 - \mathrm{F1} = R/(\eta \cdot \mathrm{TPR} + (1-\eta)(1 - \mathrm{FPR})) \leq R/(\eta(1 - R))$. $\square$

4.2 Theorem 5 Proof

Theorem 5 follows directly from the standard PAC-Bayes bound (McAllester, 1999) applied to the 0-1 loss $\ell_\theta \in [0, 1]$.

Standard PAC-Bayes bound (McAllester, 1999, Theorem 2): For any distribution $\mathcal{D}$, any hypothesis class $\mathcal{H}$, any prior P over $\mathcal{H}$ independent of the training data, and any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $S \sim \mathcal{D}^N$, for all posteriors Q over $\mathcal{H}$:

$$\mathbb{E}_{h \sim Q}[R(h)] \leq \mathbb{E}_{h \sim Q}[\hat{R}_S(h)] + \sqrt{\frac{\mathrm{KL}(Q \parallel P) + \log \frac{2\sqrt{N}}{\delta}}{2N}}$$

Application: Take $\mathcal{H} = \{h_\theta : \theta \in \Theta\}$ as the detector family, $S = V$ as the validation set, $R(h_\theta) = R(\theta)$ as the true detection error, and $\hat{R}_V(h_\theta) = \hat{R}_V(\theta)$ as the empirical detection error. The loss is bounded in $[0, 1]$ since it is the 0-1 indicator. Applying the standard PAC-Bayes bound yields the result. $\square$

4.3 Corollary 5.1 Proof

From Lemma 4, for any $\theta \in \Theta$ with $R(\theta) \leq 1/2$:

$$1 - \mathrm{F1}(\theta) \leq \frac{2R(\theta)}{\eta}$$

Taking expectations under Q:

$$\mathbb{E}_{\theta \sim Q}[1 - \mathrm{F1}(\theta)] \leq \frac{2}{\eta} \cdot \mathbb{E}_{\theta \sim Q}[R(\theta)]$$

By Jensen's inequality (since $1 - \text{F1}$ is convex in R for small R, but here we use linearity of expectation):

$$\mathbb{E}_{\theta \sim Q}[1 - \text{F1}(\theta)] = 1 - \mathbb{E}_{\theta \sim Q}[\text{F1}(\theta)]$$

Let $\text{F1}(Q) = \mathbb{E}_{\theta \sim Q}[\text{F1}(\theta)]$. Then:

$$1 - \text{F1}(Q) \leq \frac{2}{\eta} \cdot R(Q)$$

Similarly, for the empirical quantities:

$$1 - \widehat{\text{F1}}(Q) \leq \frac{2}{\eta} \cdot \hat{R}_V(Q)$$

where $\widehat{\text{F1}}(Q) = \mathbb{E}_{\theta \sim Q}[\widehat{\text{F1}}_V(\theta)]$ and $\widehat{\text{F1}}_V(\theta)$ is the empirical F1 on V.

Applying Theorem 5:

$$\begin{aligned}
1 - \text{F1}(Q) &\leq \frac{2}{\eta} \cdot R(Q) \\
&\leq \frac{2}{\eta} \left[\hat{R}_V(Q) + \sqrt{\frac{\text{KL}(Q \parallel P) + \log \frac{2\sqrt{N}}{\delta}}{2N}} \right] \\
&\leq (1 - \widehat{\text{F1}}(Q)) + \frac{2}{\eta} \sqrt{\frac{\text{KL}(Q \parallel P) + \log \frac{2\sqrt{N}}{\delta}}{2N}}
\end{aligned}$$

Rearranging:

$$\text{F1}(Q) \geq \widehat{\text{F1}}(Q) - \frac{2}{\eta} \sqrt{\frac{\text{KL}(Q \parallel P) + \log \frac{2\sqrt{N}}{\delta}}{2N}}$$

With probability $\geq 1 - \delta$. $\square$

4.4 Corollary 5.2 Proof

When $Q = \delta_{\hat{\theta}}$ is a Dirac distribution and P is uniform over a discretized grid of T thresholds:

$$\mathrm{KL}(\delta_{\hat{\theta}} \parallel P) = \int \log \frac{d\delta_{\hat{\theta}}}{dP}\, d\delta_{\hat{\theta}} = \log \frac{1}{P(\hat{\theta})} = \log T$$

Substituting into Corollary 5.1 with $|\Theta| = T$ yields the result. $\square$

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5. Connection to Paper 3 Framework

5.1 Where It Fits

This result corresponds to **Theorem 5** in the Paper 3 framework (Section 4.2, "PAC-Bayes Generalization Bound"). The framework's novelty claim #4 states:

> **Generalized Dawid-Skene equivalence**: We prove that SCX weighting reduces to Dawid-Skene weighting under trivial state partition, and that the improvement is lower-bounded by the mutual information $I(S; W)$ between states and optimal weights.

The PAC-Bayes bound extends this by providing a fully empirical performance guarantee.

5.2 How It Extends Theorem 1

Aspect	Theorem 1 (Current)	Theorem 5 (PAC-Bayes)
μ_s knowledge	Assumed known	Estimated from n_s samples
Threshold selection	Requires μ_s to set θ^*	Data-driven via empirical risk minimization
Bound type	Population	Finite-sample, high-probability
Dependence on M	$\exp(-2M\Delta_s^2)$	Captured through $\hat{R}_V(\theta)$
Additional penalty	None	$\sqrt{(\mathrm{KL}(Q\|P) + \log(1/\delta))/(2N)}$

5.3 Synergy with Corollary 4 (Finite-Sample Correction)

The current Corollary 4 already provides a naive finite-sample correction via Hoeffding:

$$|\hat{\mu}_s - \mu_s| \leq B\sqrt{\frac{\log(2M/\delta_0)}{2n_s}}$$

The PAC-Bayes approach improves on this in three ways: 1. **No union bound**: The KL term automatically accounts for the complexity of multiple states 2. **Tighter constants**: The $\sqrt{\mathrm{KL}/N}$ term is often smaller than the sum of per-state Hoeffding bounds 3. **Posterior flexibility**: We can choose Q adaptively (e.g., concentrate on the empirically best threshold) and the bound adapts

5.4 Practical Significance

The PAC-Bayes bound tells SCX practitioners:

> "Given N validation samples with known clean/noise labels, with confidence $1-\delta$, the true F1 is at least the validation F1 minus a penalty that grows as $\sqrt{(\text{number of candidate thresholds evaluated})/N}$."

This means: - For $N = 10{,}000$, $\delta = 0.05$, $|\Theta| = 100$ thresholds: penalty $\approx \frac{2}{\eta}\sqrt{(\log 100 + \log(2\sqrt{10000}/0.05))/20000} \approx \frac{2}{\eta} \cdot 0.029$ - Even with $\eta = 0.1$, the penalty is ≈ 0.58 — why so large? Because F1 is ill-defined near $\eta = 0$. For practical noise rates $\eta \geq 0.05$, the bound becomes useful.

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6. Practical Implications

6.1 How to Use the Bound

Step 1: Prepare a validation set V with known clean/noise labels (e.g., inject synthetic noise in a held-out portion of clean data).

Step 2: Define a grid of candidate thresholds $\Theta = \{\theta_1, \ldots, \theta_T\}$.

Step 3: For each $\theta \in \Theta$, compute the empirical F1 on V:

$$\widehat{\mathrm{F1}}_V(\theta) = \frac{2 \cdot \mathrm{TP}_V(\theta)}{2 \cdot \mathrm{TP}_V(\theta) + \mathrm{FP}_V(\theta) + \mathrm{FN}_V(\theta)}$$

Step 4: Choose the threshold $\hat{\theta}$ that maximizes $\widehat{\mathrm{F1}}_V(\theta)$.

Step 5: Apply Corollary 5.2 to obtain a lower bound on the true F1:

$$\mathrm{F1}(\hat{\theta}) \geq \widehat{\mathrm{F1}}_V(\hat{\theta}) - \frac{2}{\eta}\sqrt{\frac{\log T + \log \frac{2\sqrt{N}}{0.05}}{2N}}$$

with 95% confidence.

6.2 When the Bound is Tight

The PAC-Bayes bound is tight when: 1. N **is large** relative to the complexity $\mathrm{KL}(Q\|P)$ 2. **The empirical F1 is accurate** (validation set is representative of the true distribution) 3. $R(\theta) \ll 1/2$ so the F1-risk conversion is efficient

6.3 When the Bound is Loose

The bound becomes loose when: 1. N **is small** (< 100 samples per state): the complexity penalty dominates 2. η **is very small** (< 0.01): the $1/\eta$ amplification makes the bound vacuous 3. **The empirical F1 is near 0 or 1**: the 0-1 loss PAC-Bayes bound doesn't leverage the variance at the extremes (use the empirical Bernstein variant in Item 3 instead)

6.4 Relation to Catoni's PAC-Bayes

Catoni (2007) provides a refined PAC-Bayes bound with sharper constants. The bound above uses McAllester's original formulation for simplicity. In the actual paper, we can use Catoni's bound:

$$R(Q) \leq \hat{R}_V(Q) + \frac{\mathrm{KL}(Q \parallel P) + \log(1/\delta)}{\lambda N} + \frac{\lambda}{8N}$$

where $\lambda > 0$ is an inverse-temperature parameter. Optimizing over λ gives a tighter bound, especially when $\hat{R}_V(Q)$ is small.

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7. Open Questions and Extensions

1. **Data-dependent prior**: Can we use a prior that depends on the structure of the state partition? The current bound requires P to be independent of V.

2. **PAC-Bayes with F1 directly**: The standard PAC-Bayes bound applies to the 0-1 loss, not to the F1 score directly. A PAC-Bayes bound for the F1 itself would be tighter but requires more sophisticated tools (e.g., the F1's difference-of-convex structure).

3. **Unlabeled validation set**: In the pure SCX setting, we may not have ground-truth labels for the validation set. Can we bound the detector's performance using only clean validation samples? This would require a different approach — perhaps using the estimated $\hat{\mu}_s$ directly and propagating uncertainty.

4. **Semi-supervised PAC-Bayes**: Extend to settings where only a subset of the validation set has known labels (active learning for threshold calibration).

References

1. McAllester, D. A. (1999). PAC-Bayesian model averaging. *Proceedings of the Twelfth Annual Conference on Computational Learning Theory.*

2. Catoni, O. (2007). PAC-Bayesian supervised classification: The thermodynamics of statistical learning. *IMS Lecture Notes Monograph Series*, 56.

3. Germain, P., Bach, F., Lacoste, A., & Marchand, M. (2009). PAC-Bayesian learning of linear classifiers. *Proceedings of ICML.*

4. Theorem 1 (SCX Noise Detection). `../theorems/01_noise_detection_guarantee.md`

5. Corollary 4 (Finite-Sample Correction). `../theorems/01_noise_detection_guarantee.md#`